

Complex Analysis, final exam

June 12, 2023

1. Consider the elliptic curve $y^2 = 4x^3 - 8x$, $(x, y) \in \mathbb{C} \times \mathbb{C}$, and two points $(x_1, y_1) = (2, 4)$ and $(x_2, y_2) = (-1, 2)$ on the elliptic curve. Let $+$ be the addition (for points on the corresponding curve). Find out $(2, 4) + (-1, 2)$.
2. We know that Weierstrass $\wp$ -function satisfies $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$. Derive an equation relating $\wp^{(4)}$ to $\wp$, where $\wp^{(4)}$ is the fourth order derivative of $\wp$.
3. Describe the image of the upper half plane under the conformal mapping $F(z) = \int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(\zeta-\lambda)}}$ for $\lambda \in \mathbb{R}$ and $\lambda \neq -1$.
4. Suppose $f: \mathbb{C} \rightarrow \mathbb{C}$ is a holomorphic function such that the function $z \mapsto g(z) = f(z)f(1/z)$ is bounded on $\mathbb{C} \setminus \{0\}$. Show that
- (1) if $f(0) \neq 0$, then f is constant.
 - (2) if $f(0) = 0$, then there exist $n \in \mathbb{N}$ and $a \in \mathbb{C}$ such that $f(z) = az^n$ for all $z \in \mathbb{C}$.
5. Consider equation $z^2(z+1)w'' - zw' + \frac{1}{4}w = 0$. It has two solutions of the form $z^{\alpha_1}g_1(z)$ and $z^{\alpha_2}g_2(z)$ in a neighborhood of 0, where g_1 and g_2 are holomorphic. Determine α_1 and α_2 .
6. Let $g: \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C} \setminus \{0\}$ be an injective holomorphic function. Prove that g does not have an essential singularity at ∞ .
7. For simplicity, we consider elliptic functions satisfying the doubly periodic conditions $f(z+2\pi) = f(z)$, $f(z+i) = f(z)$ for all $z \in \mathbb{C}$. We have proved in class that f has the same number of zeros and poles and the sum of the zeros and that of the poles are the same modulo the lattice. Let us now prove the converse: for any given complex numbers $a_1, \dots, a_k$ and $b_1, \dots, b_k$ in the fundamental parallelogram satisfying $\sum_{i=1}^k a_i = \sum_{i=1}^k b_i + m2\pi + ni$, for some $m, n \in \mathbb{Z}$, there exists an elliptic function f that has $a_1, \dots, a_k$ as zeros and $b_1, \dots, b_k$ as poles. Hint: use θ function.
8. Let $\Omega = \mathbb{C} \setminus (-\infty, 0]$ and let $\log: \Omega \rightarrow \mathbb{C}$ be the branch of the complex logarithm on Ω that is real on the positive real axis. Show that for $0 < t < \infty$, the number of solutions $z \in \Omega$ to the equation

$$\log z = \frac{t}{z}$$



is finite and independent of t .

9. Let $\Omega_1 \subseteq \Omega_2$ be two bounded simply-connected domains in $\mathbb{C}$. We also assume that $0 \in \Omega_1$. Now suppose $f_1 : \mathbb{D} \rightarrow \Omega_1$ and $f_2 : \mathbb{D} \rightarrow \Omega_2$ are Riemann mappings, satisfying $f_1(0) = f_2(0) = 0$. Show that

$$|f_1'(0)| \leq |f_2'(0)|.$$

10. Evaluate

$$\lim_{R \rightarrow \infty} \int_0^R \sin(t^2) dt$$

assuming that the integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ is known. Hint: consider the contour

$$[0, R] \cup \{Re^{i\theta} \mid 0 \leq \theta \leq \frac{1}{4}\pi\} \cup [0, Re^{i\pi/4}].$$

